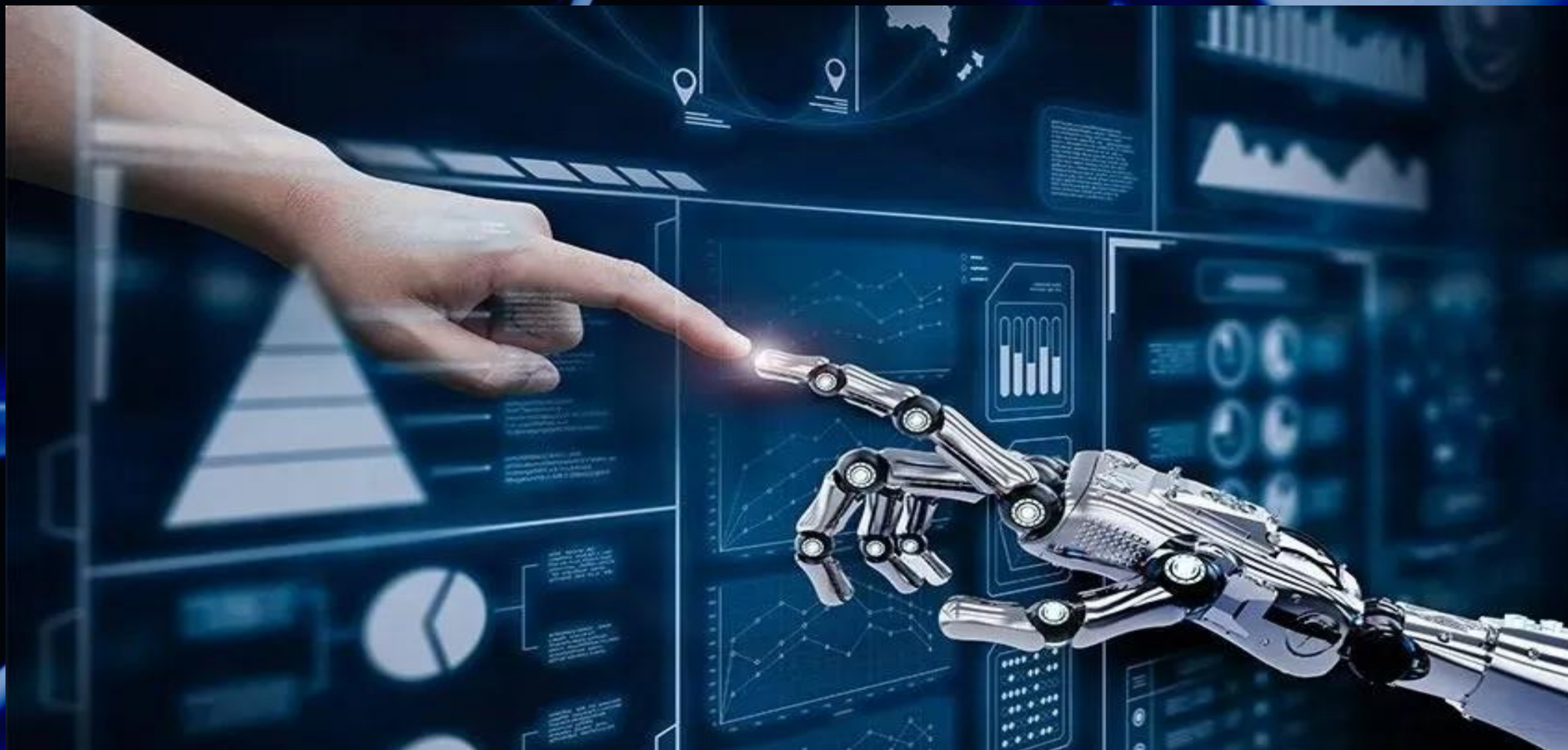


Computer Science Engineering



Engineering Study



What is Engineering?



Engineering is the **application of scientific, mathematical, and empirical knowledge** to design, build, and optimize solutions that address real-world problems and human needs.

- ❏ The word "Engineering" originates from the Latin *ingenium* — meaning cleverness or ingenuity. Engineers transform ideas into reality: they design bridges, build software systems, develop medicines, and power cities — all by applying structured problem-solving to serve society.

Who Discovers vs. Who Builds

The Scientist

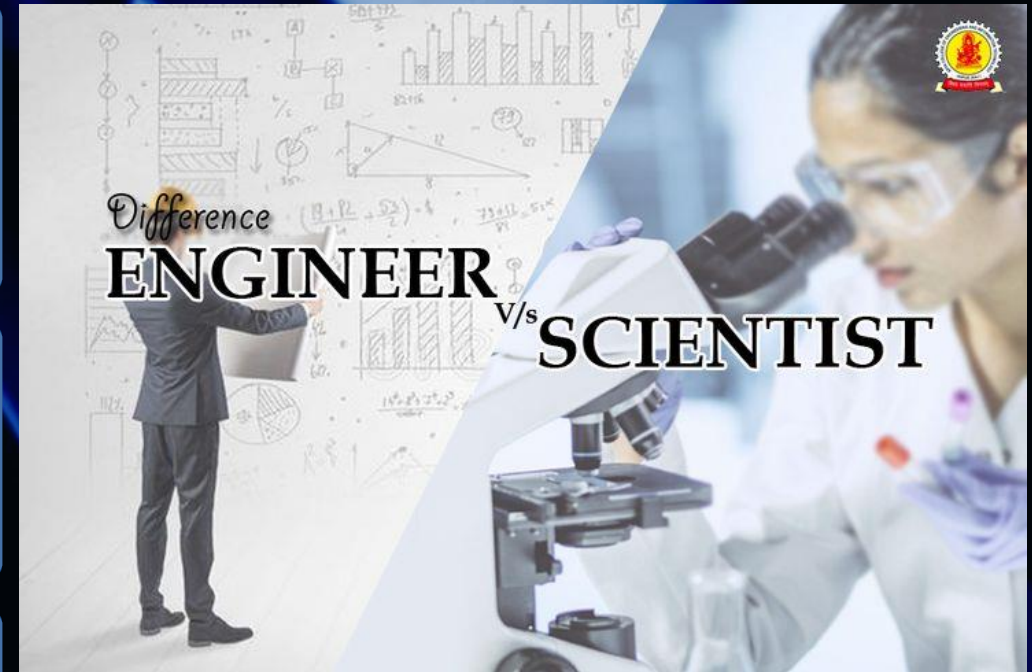
Scientists ask **"Why does this happen?"** — they investigate natural phenomena, formulate hypotheses, and pursue knowledge for its own sake. Their goal is to **understand the universe** through observation and experimentation.

The Engineer

Engineers ask **"How can we use this?"** — they apply existing knowledge to design systems, solve problems, and build products. Their goal is to create **practical, reliable, and cost-effective solutions** for society.

Where They Meet

Both roles are deeply interdependent. Scientists discover transistors; engineers build microchips. Scientists study electromagnetism; engineers design 5G networks. **Discovery fuels design, and design fuels discovery.**



Two Sides of the Same Coin

Science and Engineering are distinct yet complementary disciplines that together drive human progress.



Science

Focuses on **discovery and understanding**. Uses the scientific method — observe, hypothesize, test. Output is knowledge, theories, and published research. Success is measured by **truth and accuracy**.



Engineering

Focuses on **design and creation**. Uses design thinking — define, prototype, optimize. Output is products, systems, and processes. Success is measured by **functionality, efficiency, and cost**.



Key Difference: Constraints

Science seeks an **unconstrained truth**. Engineering always works within constraints — time, budget, materials, safety, and human usability. An engineer's best solution is the **optimal one within limits**.

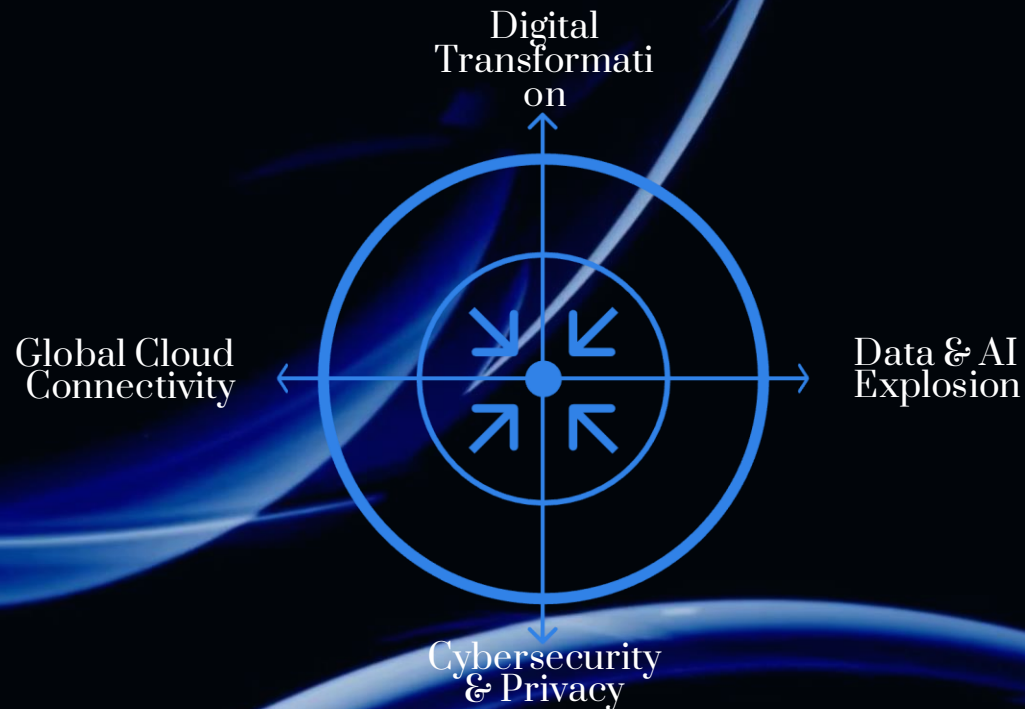


Mutual Dependency

Engineering advances create better scientific instruments (e.g., particle accelerators). Scientific breakthroughs enable new engineering fields (e.g., quantum computing). They **co-evolve in a feedback loop**.

TODAY'S WORLD

Needs & Wants of CSE in the 21st Century



CSE engineers are the architects of the digital age — every sector from healthcare to agriculture now depends on software and data systems.

Critical Demands

- 1 **AI & Machine Learning** — automating decisions, diagnostics, and discovery across every industry
- 2 **Cloud Computing** — on-demand scalable infrastructure powering global services
- 3 **Cybersecurity** — protecting critical data, infrastructure, and national systems
- 4 **IoT & Embedded Systems** — connecting the physical and digital world
- 5 **Software for Social Good** — ed-tech, health-tech, agri-tech, and fintech

The world doesn't just *want* CSE — it fundamentally **cannot function** without it.

Robotic Surgery System

Computer Science Engineering (CSE)

Role: Software, AI, robot control, and data management.

Mechanical Engineering

Role: Designs robotic arms and movement.

Electronics & Communication Engineering (ECE)

Role: Develops sensors, cameras, and communication.

Electrical & Electronics Engineering (EEE)

Role: Provides power and controls motors.

Biomedical Engineering

Role: Designs surgical tools and ensures patient safety.

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nd



✦ HOW THE DISCIPLINES COMBINE: ✦

- ★ **CSE (Software/AI)** – Motion-scaling algorithms translate the surgeon's hand movements into precise robotic arm motion; tremor filtering removes natural hand shake; computer vision processes 3D endoscopic camera feed in real time.
- ★ **Biomedical Engineering** – Designs surgical instruments (graspers, scissors, cauterizers) to be biocompatible and safe for internal tissue contact.
- ★ **Mechanical Engineering** – Builds robotic arm joints for fine, multi-directional articulation inside a small incision.
- ★ **Electrical Engineering** – Powers the motors, sensors, and haptic feedback systems that let the surgeon "feel" resistance.

REAL-WORLD IMPACT: ✦

- ★ Smaller incisions → less blood loss, faster recovery, lower infection risk.
- ★ Higher precision → improved outcomes in complex procedures.
- ★ Surgeon operates from a console → enables remote/telesurgery research globally.



Discipline Can Decide
Your **Engineering**



Misconceptions About CSE Engineering



'CSE is just coding all day'

CSE encompasses system design, architecture, algorithms, mathematics, project management, and human-computer interaction. Coding is one skill — **not the whole discipline**. A CSE engineer also thinks, designs, communicates, and leads.



'CSE engineers are antisocial loners'

Modern CSE thrives on **teamwork, communication, and empathy**. Agile teams, open-source collaboration, client-facing roles, and product management all demand strong interpersonal skills alongside technical expertise.



'CSE has no room for creativity'

Software engineering is one of the most creative professions. Designing elegant algorithms, building intuitive interfaces, and architecting scalable systems require **imagination and innovation** at every step — not just logic.

Skills Required in CSE Engineering Today



The CSE engineer of the 21st century is a **T-shaped professional** — deep expertise in one area, broad knowledge across many.

Skill Priority in Industry Today

Skill Area

AI & Machine Learning
Cloud Computing
Cybersecurity
Data Engineering
DevOps & Automation
Communication & Lead...



Industry Demand (%)

Gamma AI Features

- ★ 1. **AI Presentation Creation** – Creates professional presentations from a simple prompt.
- ★ 2. **Smart Design** – Automatically applies attractive layouts, themes, and formatting.
- ★ 3. **AI Content Generation** – Generates titles, text, and summaries instantly.
- ★ 4. **Images & Media** – Adds AI-generated images, icons, GIFs, and videos.
- ★ 5. **Easy Editing** – Modify slides quickly using drag-and-drop tools.
- ★ 6. **Collaboration** – Multiple users can edit and comment in real time.
- ★ 7. **Export & Share** – Export presentations as PDF or PowerPoint, or share with a link.
- ★ 8. **Templates** – Provides ready-made templates for education, business, and reports.
- ★ 9. **Responsive Design** – Presentations automatically adjust to different screen sizes.



Gamma AI

Thank You

Always thank the audience for lending you their time. This part can often be found in the last slide of your accounting and finance PPT.

- *Kaviya R*